

SAYAN DUTTA

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Kolkata, West Bengal | [GITHUB](#) | [LinkedIn](#)

BS-MS student in Computer Science and Data Analytics at IIT Patna with research experience in NLP, LLM reasoning, and adaptive AI systems. Co-author of an ACL ARR 2026 submission and Research Intern at BITS Pilani. Interested in memory-augmented AI, cognitive architectures, reasoning, and continual learning.

EDUCATION

Indian Institute of Technology Patna

2025 – 2030 (Expected)

BS-MS, Computer Science and Data Analytics (Integrated 5-Year)

- CPI: 8.64 / 10.0
- Coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Optimization Fundamentals, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

RESEARCH INTERESTS

Large Language Models · Cognitive Architectures · Memory-Augmented AI · Reasoning Systems · Natural Language Processing · Continual Learning · AI Systems

RESEARCH EXPERIENCE

Research Intern — BITS Pilani (Remote)

Apr 2026 – May 2026

Project: ECOT-ERG: Empathetic Dialogue Generation with Structured Chain-of-Thought | Supervisor: Dr. Srishti Gupta, PostDoctoral Fellow, Dept. of Computer Science

- Implemented and trained the ECOT-ERG framework integrating emotion prediction, cause identification, chain-of-thought supervision, and T5-based response generation into a unified end-to-end pipeline.
- Built modular emotion prediction and cause identification components enabling independent ablation of each reasoning stage without full system retraining.
- Fine-tuned T5 models with FP16 mixed-precision training; constructed DPO preference pairs from ranked outputs for empathy alignment.
- Executed 16 experimental configurations (4-way ablation across CoT supervision, Emotion Prediction, Cause Identification, DPO); evaluated on BERTScore, EASE, Cause-F1, and Relevance-Accuracy.
- Co-authored ACL ARR May 2026 submission; conducted error analysis and generated qualitative outputs for paper discussion.

PUBLICATIONS

Towards Causally Faithful Empathetic Dialogue Generation via Structured Appraisal Reasoning

Srishti Gupta, Ayushman Sarkar, Sayan Dutta, Shashank Vishwakarma, Rishikesh Raj, Arghya Jana, Nagandla Tarun Teja

ACL ARR 2026 (Under Review)

Contributions: Implemented and trained the core ECOT-ERG architecture, developed structured reasoning-chain generation, emotion prediction and cause identification modules, executed ablation studies, and performed qualitative and quantitative evaluation.

PROJECTS

Blix - Cognitive Architecture for Adaptive AI Systems

2024 – Present

Independent Project | Python, PyTorch, FAISS, Pydantic

- Developed a modular AI architecture integrating hierarchical memory, temporal retrieval, belief tracking, and reflection mechanisms for adaptive decision-making.
- Improved knowledge consistency through truth maintenance, state-aware retrieval, and confidence estimation modules.
- Enhanced agent robustness via failure memory, self-correction, and replanning mechanisms for experience-driven adaptation.
- Established workspace-based coordination combining attention, planning, verification, and evaluation pipelines for AI systems research.
- Engineered a fully local, privacy-preserving framework optimized for consumer hardware and extensible AI systems experimentation.

ASES - AI-Driven Precision Agriculture Advisory Platform (Capstone Project)

Python, Scikit-learn, Google Gemini API, Streamlit, SQLite

- Developed crop and fertilizer recommendation modules using feature-normalized KNN models over multi-dimensional soil and environmental attributes.
- Enhanced recommendation quality by integrating LLM-based contextual advisory generation through structured prompting with Google Gemini API.
- Designed a lightweight end-to-end application with Streamlit frontend and SQLite backend for persistent storage and local deployment.
- Established a modular pipeline separating prediction, recommendation, and natural-language advisory components for extensible agricultural decision support.

TECHNICAL SKILLS

Languages: Python, C++, C, Java

ML / DL: PyTorch, Hugging Face, Transformers, Seq2Seq Trainer, Scikit-learn

LLM & Reasoning: Retrieval-Augmented Generation (RAG), FAISS, Chain-of-Thought, Direct Preference Optimization (DPO), Prompt Engineering

NLP: Sentence Transformers, Empathetic Dialogue Systems, BERTScore, EASE, Cause-F1

Engineering: Git, FastAPI, REST APIs, Pydantic, Linux, CUDA, FP16 Mixed Precision, Jupyter

Databases: MySQL, SQLite

LEADERSHIP

Lead Google Student Ambassador

2025 – Present

Indian Institute of Technology Patna

- Coordinated technical outreach initiatives serving 100+ students across AI/ML workshops and developer events.

ACHIEVEMENTS

- Co-authored an ACL ARR 2026 submission on structured chain-of-thought reasoning for empathetic dialogue generation.
- Selected as Research Intern under Dr. Srishti Gupta, Department of Computer Science, BITS Pilani.
- Lead Google Student Ambassador at IIT Patna, coordinating technical outreach initiatives serving 100+ students.
- Conducted 16 ablation experiments and contributed to DPO alignment, error analysis, and evaluation pipelines for ECOT-ERG.
- Building research-oriented AI systems focused on memory, reasoning, and adaptive architectures.